Exact carry-position identities for deep inverse prefixes

27 August 2026

For a positive odd integer x, put

$$U_+(x) := \frac{3x+1}{2^{v_2(3x+1)}}. \tag{1}$$

For a positive exponent word $E = (e_1, \ldots, e_h)$, define

$$\mathrm{B}(E) := \sum_{i=1}^{h} e_i, \qquad \mathrm{C}(E) := \sum_{i=1}^{h} 3^{h-i} 2^{e_1+\cdots+e_{i-1}}, \tag{2}$$

and

$$\mathrm{D}(E) := \mathrm{C}(E) + 2^{\mathrm{B}(E)} - 3^h. \tag{3}$$

An exact path from z to r with word E satisfies

$$2^{\mathrm{B}(E)}(r+1) = 3^h(z+1) + \mathrm{D}(E). \tag{4}$$

Write

$$e_i = 1 + c_i, \qquad c_i \geq 0, \qquad t := \sum_{i=1}^{h} c_i, \tag{5}$$

and define the cumulative excess

$$p_k := \sum_{i=1}^{k} c_i \qquad (0 \leq k \leq h). \tag{6}$$

Thus $p_0 = 0$ and $p_h = t$.

1 The two exact slack identities

For fixed h and t, the shifted-carry extrema are

$$\mathrm{D}_{\min} = 2^h(2^t - 1), \qquad \mathrm{D}_{\max} = 2\,3^{h-1}(2^t - 1). \tag{7}$$

Theorem 1 (carry-position decomposition)**.** *Every positive exponent word of length h and excess t satisfies*

$$\boxed{\frac{\mathrm{D}(E) - \mathrm{D}_{\min}}{3^h} = \frac{1}{3}\sum_{k=1}^{h-1}\left(\frac{2}{3}\right)^k (2^{p_k} - 1),} \tag{8}$$

and

$$\boxed{\frac{\mathrm{D}_{\max} - \mathrm{D}(E)}{3^h} = \frac{1}{3}\sum_{k=1}^{h-1}\left(\frac{2}{3}\right)^k (2^t - 2^{p_k}).} \tag{9}$$

Proof. Equation (2) becomes

$$\mathrm{C}(E) = \sum_{i=1}^{h} 3^{h-i}2^{i-1+p_{i-1}}. \tag{10}$$

When all excess is in the final letter, every p_{i-1} in (10) is zero. Subtracting this minimum carry from (10), and putting $k = i - 1$, gives

$$\mathrm{D}(E) - \mathrm{D}_{\min} = \sum_{k=1}^{h-1} 3^{h-k-1}2^k(2^{p_k} - 1). \tag{11}$$

Division by 3^h proves (8).

When all excess is in the first letter, every p_k, $k \geq 1$, is t. Subtracting (10) from this maximum carry gives

$$\mathrm{D}_{\max} - \mathrm{D}(E) = \sum_{k=1}^{h-1} 3^{h-k-1}2^k(2^t - 2^{p_k}). \tag{12}$$

This proves (9). □

2 The cumulative-excess squeeze

Put

$$\mathcal{A}_h(r) := \left(\frac{2}{3}\right)^h (r+1), \tag{13}$$

and define the two sharp endpoints

$$L_{h,t}(r) := 2^t \left(\mathcal{A}_h(r) - \frac{2}{3}\right) - \frac{1}{3}, \tag{14}$$

$$U_{h,t}(r) := \left(\frac{2}{3}\right)^h (2^t r + 1) - 1. \tag{15}$$

Corollary 2 (exact endpoint slack)**.** *If E is an exact path from z to r, then*

$$\boxed{U_{h,t}(r) - z = \frac{1}{3}\sum_{k=1}^{h-1} \left(\frac{2}{3}\right)^k (2^{p_k} - 1),} \tag{16}$$

$$\boxed{z - L_{h,t}(r) = \frac{1}{3}\sum_{k=1}^{h-1} \left(\frac{2}{3}\right)^k (2^t - 2^{p_k}).} \tag{17}$$

Proof. Equation (4) gives

$$z + 1 = 2^t \mathcal{A}_h(r) - \frac{\mathrm{D}(E)}{3^h}.$$

Subtract the identities obtained by replacing $\mathrm{D}(E)$ with the two extrema in (7), and apply Theorem 1. □

Corollary 3 (two-sided position squeeze)**.** *For every $1 \le k < h$,*

$$\boxed{\frac{2}{3}\left(1 - \left(\frac{2}{3}\right)^k\right)(2^t - 2^{p_k}) \le z - L_{h,t}(r),} \tag{18}$$

$$\boxed{\left(\frac{2}{3}\right)^k \left(1 - \left(\frac{2}{3}\right)^{h-k}\right)(2^{p_k} - 1) \le U_{h,t}(r) - z.} \tag{19}$$

Proof. The sequence p_i is nondecreasing. In (17), retain the terms $1 \le i \le k$ and use

$$2^t - 2^{p_i} \ge 2^t - 2^{p_k}.$$

The resulting geometric sum is (18). In (16), retain the terms $k \le i < h$ and use

$$2^{p_i} - 1 \ge 2^{p_k} - 1.$$

This gives (19). □

Corollary 4 (finite screen for a bounded start)**.** *Suppose that* $1 \le z < N$. *For every* $1 \le k < h$, *put*

$$\alpha_k := \frac{2}{3}\left(1 - \left(\frac{2}{3}\right)^k\right), \qquad \beta_k := \left(\frac{2}{3}\right)^k \left(1 - \left(\frac{2}{3}\right)^{h-k}\right). \tag{20}$$

Then the integer p_k *must satisfy*

$$\boxed{2^{p_k} > 2^t - \frac{N - L_{h,t}(r)}{\alpha_k},} \tag{21}$$

$$\boxed{2^{p_k} \le 1 + \frac{U_{h,t}(r) - 1}{\beta_k}.} \tag{22}$$

If these two bounds contain no power of two with exponent in $\{0, 1, \ldots, t\}$, *then no such prefix exists.*

Proof. Use $z < N$ in (18) and $z \ge 1$ in (19), and solve for 2^{p_k}. Since p_k is an integer between zero and t, the last assertion follows. □

The sharp all-excess interval uses only t. Equations (16)–(22) retain where that excess occurs. They give an exact finite obstruction at every proposed triple (h, t, r), but they do not yet exclude all deep triples simultaneously.